Reinforcement Learning based Controller for Grid-Connected PUC PV Inverter

Alamera Nouran Alquennah
Department of Electrical and Computer Engineering
Texas A&M University
College Station, TX, USA
Texas A&M University at Qatar
Qatar Foundation, Doha, Qatar
alamera_nq@tamu.edu

Melanie Chida
Department of Statistics
Texas A&M University
College Station, TX, USA
melliechida@tamu.edu

Tassneem Zamzam
Department of Electrical and Computer Engineering
Texas A&M University
College Station, TX, USA
Texas A&M University at Qatar
Qatar Foundation, Doha, Qatar
tzamzam@tamu.edu

Mohamed Trabelsi
Electronics and Communications Engineering Department
Kuwait College of Science and Technology
Kuwait City, Kuwait
m.trabelsi@kcst.edu.kw

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***Abstract*—Packed-U-Cell (PUC) is a single DC source multi-level inverter that can be used in many applications such as grid-connected photovoltaic (PV) systems. In this application, the total harmonic distortion (THD) of the generated current signal should be minimized while operating at the unity power factor (maximum active power transfer). These objectives can be achieved by regulating the auxiliary capacitor voltage around its reference value while tracking the reference current signal that varies with the PV maximum power point (MPP). Thus, this paper proposes a reinforcement learning (RL) based controller that satisfies the aforementioned control objectives using the actor-critic RL architecture and the proximal policy optimization (PPO) learning algorithm. The designed RL-based controller is applied on a single-phase 5-level PUC inverter. The proposed design is validated through simulations where the obtained control policy resulted in a maximum absolute voltage error of 1.9 V and a THD value of 2% and 4.5% for reference current peak values of 8.8A and 4.2A, respectively. Furthermore, the proposed RL-based controller shows high robustness to parameter variations (different capacitor and inductor sizes).**

***Index Terms*—Packed-U-Cell, Multilevel Inverter, PV, Reinforcement Learning, Grid Connection**

I. Introduction

The generated power from PV plants has increased annually in 2022, where around 220 GW of DC power was installed globally and it is expected to reach 356 GW by 2026 [1]. This has led to an increasing need for high-efficiency inverters in grid-connected applications. Multi-level inverters (MLIs), compared to the conventional 2-level inverters, are suitable for such systems due to their ability to generate higher power quality signals. This is in addition to the lower harmonic polluted signals, lower electromagnetic interference, and reduced voltage stress on the power semiconductor devices [2].

Packed-U-cell (PUC) is a single DC source MLI (SDC-MLI) that was proposed in 2011 [3] and has grabbed since then the attention of many PV research studies [4]–[7]. The PUC topology can generate 5 or 7 voltage levels at the output terminals using only one DC source, one flying capacitor, and six power switches (reduced number of passive/active components compared to many MLIs typologies) [8]. The number of voltage levels and the quality of the current signal fed to the grid depends on the control algorithm applied to the PUC-MLI. The controller function mainly focuses on regulating the capacitor voltage around a desired value and tracking the reference current. The capacitor's voltage is regulated around 1/3 or 1/2 of the input DC voltage to generate 7 or 5 voltage levels, respectively. The current signal quality has higher priority in grid-tied systems to satisfy the IEEE 929-2000 standard where the THD should be less than 5%.

To satisfy the operation requirements, several controllers were designed for PUC-MLI as summarized in [9] including cascaded PI controller [10], multi-band Hysteresis [11], finite control set-model predictive control FCS-MPC [4], Lyapunov based controller [12], and sliding mode controller (SMC) [13]. However, these control techniques are optimized for a specific operating condition and tested for different scenarios as in [4]. To accommodate the stochastic operating conditions of PV systems, those controllers were modified to be auto-tuned using artificial neural network (ANN) [14], [15], polynomial-based weighting factors [16], and fuzzy tuning approach [17]. In addition, ANN-based controllers were proposed such as in [18] and [19].

The performance of the model-based control techniques, such as SMC and MPC, depends on the accuracy of the mathematical model used in the control design. This is in addition to the requirement of solving an online optimization problem at each time step. On the other hand, the ANN-based controllers require collecting data using another control technique to train the ANN. This process can be performed online or offline. After the training and testing stages, the ANN replaces the base controller in the system. In Reinforcement learning (RL) based controllers, the aforementioned design requirements are tackled where the controller is designed without requiring the knowledge of the system's model or the need for collecting data from the operating system or environment. The controller (agent) learns the appropriate control policy through its interaction with the environment directly considering its dynamic operating states. Compared to FCS-MPC and Lyapunov-based MPC techniques, in RL-based controller (RLC), the implemented trained agent selects

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the appropriate action by mapping the state space to the action space using the learned policy without the need to solve an optimization problem. The importance of this advantage increases with the increase in the number of the switching configuration resulting in decreasing the online computation burden. Furthermore, since RLC is a model-free controller, its performance is robust against the change in the component size such as the inductor and capacitor. However, these advantages of RLC come with the need to accommodate the various operation conditions in the designed training environment. In addition, the design of the reward function is problem-dependent and it significantly affects the learning performance of the RLC agent [20].

In [21], an RL approach was used to auto-tune FCS-MPC for a three-phase stand-alone inverter. Also, it was applied in the context of balancing distribution grids based PV [22] and frequency control [23], and extracting MPPT in grid-tied PV system [24]. More details about this research aspect can be found in [25]. In [26], the actor-critic RL architecture using the proximal policy optimization (PPO) algorithm was used to stabilize a buck-boost converter in a DC microgrid application.

In the context of grid-connected PV systems, RLC is a suitable control algorithm that considers the stochasticity of the environment where the PV DC voltage and current vary with the weather and operating conditions. Therefore, this paper proposes an RLC using the actor-critic architecture and the PPO learning algorithm for a single-phase grid-tied PV PUC-5 inverter to generate current with a unity power factor. It is worth mentioning that the complexity of the action space and the training process of RLC increase with the number of switches and switching states. This implies that the action space and the training process of MLIs are more complex than the conventional DC-AC inverters. However, this paper shows that the proposed RL-based control design applied could successfully achieve the control objectives of the PUC-5 MLI.

The next section details the PUC topology and the grid-tied system followed by the RLC design and its learning algorithm. The proposed RLC is verified through simulations and the results are shown to prove its applicability for MLIs.

II. PUC Topology and Modelling

The PUC-MLI topology under study consists of 3 cells connected to each other through two flying capacitors, C_1 and C_2, as shown in Fig. 1. The PV system is connected to the upper capacitor through a DC-DC converter that regulates C_1 voltage (V_1) at its desired value (V_1^*) through the PV maximum power point tracking (MPPT) algorithm.

The voltage levels at the output terminals, V_{AN}, are formed from the capacitors' voltages V_1 and V_2 by the six power switches as shown in Table I. Each cell has two switches operated in a complementary manner that when one of them is ON (1) the other should be OFF (0).

To have symmetric voltage levels at the output terminals, a controller is needed to regulate V_2 at its desired value, V_2^*. In this study, V_2^* is chosen to be $V_1/2$ to generate 5 DC voltage levels.

The inverter injects the generated AC current, i_g, to the grid through a filtering inductor, L_f. The maximum active power is extracted from the PV system by synchronizing i_g with the grid voltage, v_g, aiming for a unity power factor.

The mathematical model of the system's topology shown in Fig. 1 can be formed as following:

$$\frac{dV_2}{dt} = \frac{1}{C_2}(s_3 - s_2)i_g(t) \tag{1}$$

$$\frac{di_g}{dt} = \frac{1}{L_f}(V_{AN}(t) - v_g(t)) \tag{2}$$

$$V_{AN}(t) = (s_1 - s_2)V_1(t) + (s_2 - s_3)V_2(t) \tag{3}$$

Considering the sampling time T_s at which the states are measured and the switches are controlled, the model in (1)-(2) can be discretized as shown in (4)-(5), where k refers to the sample number.

$$V_2(k+1) = (s_3 - s_2)\frac{T_s}{C_2}i_g(k) + V_2(k) \tag{4}$$

$$i_g(k+1) = (V_{AN}(k) - v_g(k))\frac{T_s}{L_f} + i_g(k) \tag{5}$$

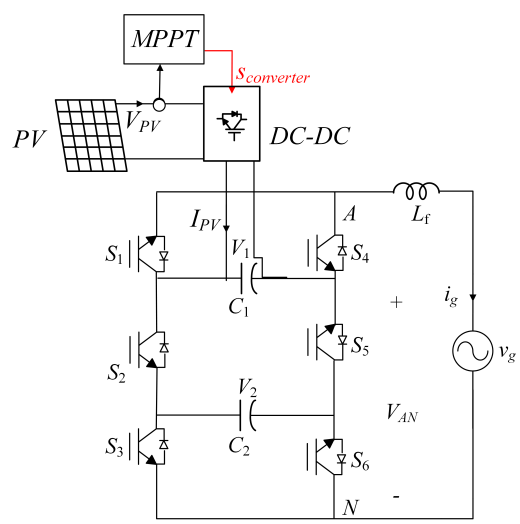

Fig. 1. The topology of 5-levels Packed-U-Cells inverter based grid-connected PV system.

III. RL Based Control Design

The framework of the proposed RLC is illustrated in Fig.2. Its function is to choose the switching configuration at each time step satisfying the following objectives:

1) Tracking the current reference signal at each time step to minimize the total harmonic distortion.
2) Regulate the capacitor voltage V_2 around its reference value V_2^*.

Based on the control objectives and the environment of the problem under study, the action space, the state space, and the reward function are defined as follows:

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TABLE I
SWITCHING PATTERNS, THE OUTPUT VOLTAGE LEVELS AND THE ACTION SPACE

S_1	S_2	S_3	V_{AN}	**Action Space**
0	0	0	0	a_{s_1}
0	0	1	$-V_2$	a_{s_2}
0	1	0	$-V_1+V_2$	a_{s_3}
0	1	1	$-V_1$	a_{s_4}
1	0	0	V_1	a_{s_5}
1	0	1	V_1-V_2	a_{s_6}
1	1	0	V_2	a_{s_7}
1	1	1	0	a_{s_8}

A. Action Space

The switching configurations are the action space as shown in Table I. There are 8 switching configurations, and at each time step (k), the RL agent decides the action based on the measured states. Thus, the action space is discrete.

B. State Space

At each time step k, the following values are measured from the environment: V_{1k}, V_{2k}, i_{gk}, v_{gk}, and θ_k which is the phase angle of v_g obtained from a phase-locked loop (PLL) circuit as shown in Fig. 1. This is in addition to the DC-DC converter output current (I_{PV}). These measurements are used to calculate the reference current peak (I^*_{pk}), the current reference value (i^*_{gk}), and the voltage reference value (V^*_{2k}) at time step k, as following:

$$I^*_{pk} = \sqrt{2} I_{PV} \tag{6}$$

$$i^*_{gk} = I^*_{pk} sin(\theta_g) \tag{7}$$

$$V^*_{2k} = V_{1k}/2 \tag{8}$$

The state space is chosen to be: V_{1k}, V_{2k}, i_{gk}, v_{gk}, i^*_{gk} and I^*_{pk}. Therefore, the state space is a 6-dimension continuous space. These states measured at state k will be the input to the RL agent to decide the required action to state $k+1$.

C. Reward function

The objective is to track the reference signals of the capacitor voltage and grid current, $V^*_2(t)$ and $i^*_g(t)$, respectively. The deviation of the measured current i_{gk} and the voltage state V_{2k} from their desired values at time step k are defined as following:

$$e_{ik} = ||i^*_{gk} - i_{gk}|| \tag{9}$$

$$e_{vk} = ||V^*_{2k} - V_{2k}|| \tag{10}$$

Also, two penalty terms are defined to bound the current and voltage errors as follows:

$$p_{ik} = w_i(u[i_{gk} - (1+I_l)I^*_{pk}] + u[-(i_{gk} + (1+I_l)I^*_{pk})]) \tag{11}$$

$$p_{vk} = w_v(u[V_{2k} - (1+V_l)V^*_2] + u[V_{2k} - (1-V_l)V^*_2]) \tag{12}$$

where p_{ik} and p_{vk} are the current and voltage penalties, respectively. w_i and w_v are weighting factors, I_l is the current bounding limit, V_l is the voltage bounding limit and $u[.]$ is unit step function.

The reward function is designed to be a function of the obtained errors and penalties at step k after taking the k-step action (a_{sk}). Hence, the reward at step k (r_k) can be written as follows:

$$r_k = -e_{ik} - \alpha e_{vk} - p_{ik} - p_{vk} \tag{13}$$

where α is a weighting coefficient to prioritize the control objectives as shown in [14] and [15]. The objective of the RL agent is to maximize the obtained reward which will lead to minimizing the errors and penalties.

D. RL Architecture and Learning Algorithm

The actor-critic RL architecture was used in the RLC design which is a combination of the policy-based and value-based methods. The critic network is designed to have one hidden layer with 30 neurons and $tanh$ activation function. The actor-network is designed to have one hidden layer with 20 neurons and rectified linear unit (ReLU) activation function. Since the action space is discrete, the $softmax$ function is applied to the output layer. Additionally, the $Adam$ optimization algorithm is chosen for both the actor and critic networks. Lastly, for this study, the PPO learning algorithm is used.

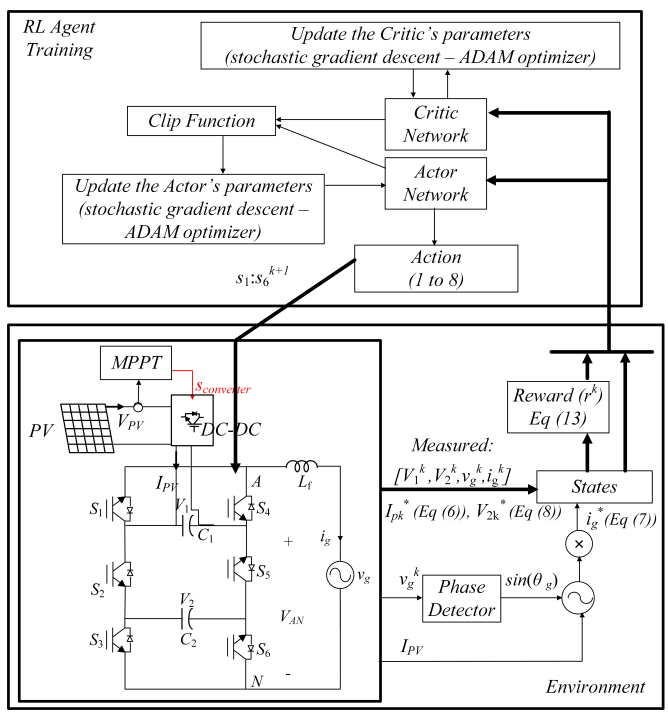

Fig. 2. The training process of the proposed RL-based controller applied on PUC5-PV based application.

IV. SIMULATION RESULTS AND DISCUSSION

The PV system was designed and simulated using SAM software (system advisor model powered by NREL). The 30-minute weather data of College Station - TX, USA from NREL solar resource library was used for this study. The design consists of a string of three SunPower SPR-440NE Mono-cSi modules with 6.04A and 72.9V maximum power point

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current and voltage, respectively. Ideal operating conditions were assumed for the PV system. The DC operating voltage and power were extracted to train the RLC. The best results were obtained when the RLC was trained on three different I_p^* levels which are: 8A, 5A and 3A. The RLC is then tested using data that was not part of the training conditions. Fig. 3 shows the weather conditions and the DC voltage and power for 24 hours that were used to generate the results shown in this paper. Each weather reading was fixed for 6 complete sinusoidal cycles to measure e_v and current THD.

The PUC5 model was designed using MATLAB/Simulink and the PV data were added to the model to create the operating environment as illustrated in Fig.2. The RLC was designed using MATLAB as described in section III. The design parameters of the PUC5 inverter are shown in Table II as well as the RLC parameters.

TABLE II
SYSTEM PARAMETERS

Parameters	Value	Parameters	Value
Sampling time (T_s)	$20\mu s$	I_l	0.05
Fundamental frequency	$60Hz$	V_l	0.1
Grid Voltage v_g (peak)	170 V	α	0.2
Filtering inductor L	$10mH$	w_v	0.5
PPO clipping factor	0.2	w_i	5
RL discount factor (γ)	0.95	Capacitor C_2	$1mF$
PPO mini-batch size	200	critic learning rate	0.0008
Number of steps/episode	2500	actor learning rate	0.0008

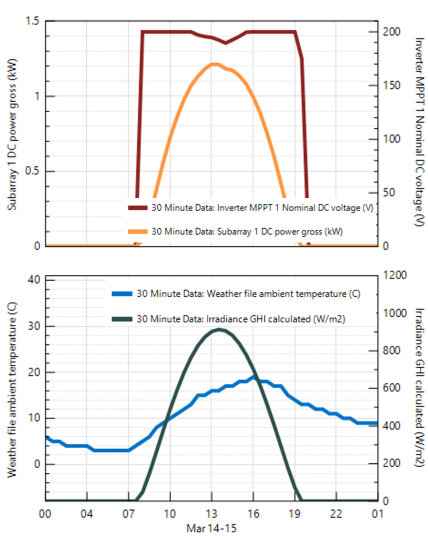

Fig. 3. The irradiance, the ambient temperature and the output DC voltage and DC power of a PV system for 24 hours.

The trained agent was evaluated using the testing case shown in Fig. 3 to validate the RLC design. The results showed that the proposed RLC regulated the capacitor voltage V_2 around its reference value V_2^* which follows the change in V_1 as shown in Fig. 4. The root mean square voltage error is found to be 1.4V for the complete testing scenario. The minimum e_v value is 0.03V and the maximum e_v value is 1.9V. As a

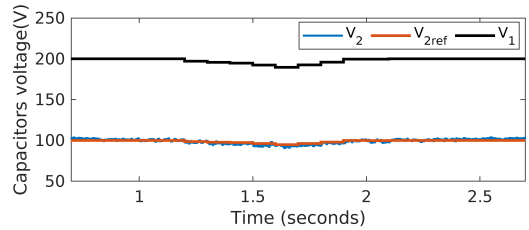

Fig. 4. The DC-DC converter output voltage V_1, the capacitor voltage V_2 and its reference signal versus time.

result of the regulated V_2 and the properly chosen actions by RLC, the 5 DC levels of the inverter voltage were generated for all operating conditions. Fig. 5 depicts an example when V_1 dropped from 192.4V to 189.5V.

It is evident from Fig. 6 that the trained agent successfully tracked the current reference signal for varying I_{PV} which ensures that the proposed RLC agent was trained to reduce e_i at each time step. This also satisfies the control objective to extract the maximum active power from the PV system by generating a synchronized current signal with grid voltage leading to a unity power factor. This synchronization is illustrated in Fig. 7. The results shown in Fig. 4 and Fig. 6 asserts the capability of the proposed RLC for PUC-MLI operating in dynamic conditions generally and PV application specifically.

The quality of the generated grid current was analyzed by calculating the total harmonic distortion (THD) for different I_{PV}. At I_p^* equals 5.2A, the obtained THD is 3.56%, and at I_p^* equals 7.7A, the THD is 2.32%. The spectrum of the two cases is presented in Fig. 8 and Fig. 9, respectively. Table III lists the obtained THD and e_v for different I_p^* and V_1 values.

In addition, the robustness of the proposed RLC against the change in C_2 and L_f sizes was investigated for the case when I_p^* equals 7A. As shown in Fig. 10, changing C_2 size from $100\mu F$ to $2000\mu F$ maintained the e_v value under $1V$ and the maximum i_g THD was 3.14% which was obtained when C_2 was $100\mu F$. The difference between the highest and lowest THD values is 0.5%. On the other hand, varying L_f affected i_g THD more where the difference between the highest and lowest obtained THD is 1.52%, as shown in Fig. 11. In both cases, the range of e_v is 0.6V.

The performance of the RLC-trained agent (Table II parameters) is compared in terms of current THD and e_v to the FCS-MPC, PI controller, online SMC (SMC1) and offline SMC (SMC2) techniques considered in [9]. The simulation current peak is set to 10A, the filtering inductor and the PUC capacitor are 5mH and 1mF, respectively. The comparison is summarized in Table IV which shows that under a fixed current peak, the obtained THD by RLC is similar to SMC1 with slight enhancement over SMC2. The e_v is higher than all the control techniques except PI controller. However, in-deep investigation is required to compare the controllers' dynamic performance, online computation burden, and switching losses.

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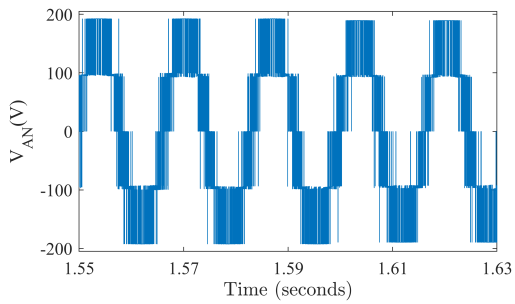

Fig. 5. The 5 DC levels of the inverter output voltage versus time using RLC controller.

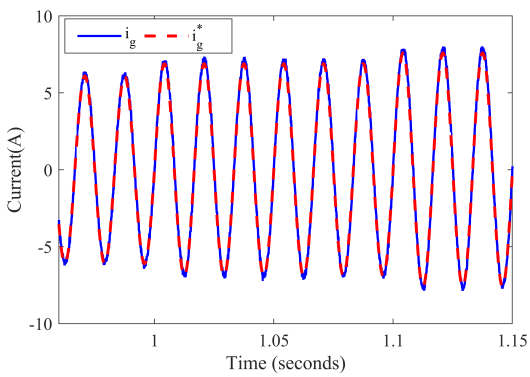

Fig. 6. The grid current i_g and its reference signal versus time.

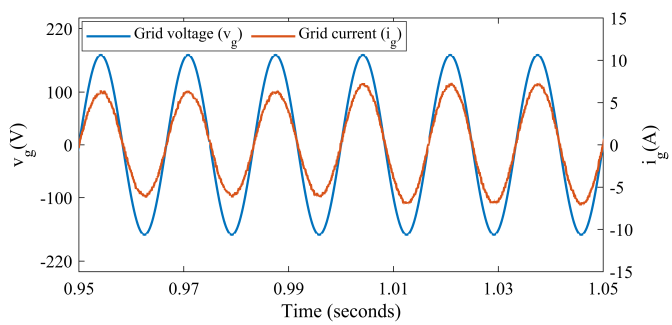

Fig. 7. The synchronization between the generated grid current i_g and the grid voltage versus time.

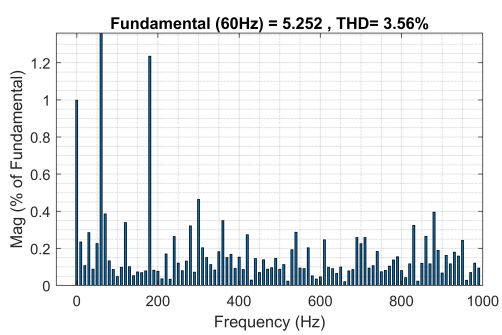

Fig. 8. The spectrum of the generated grid current i_g at 5.2A current reference peak.

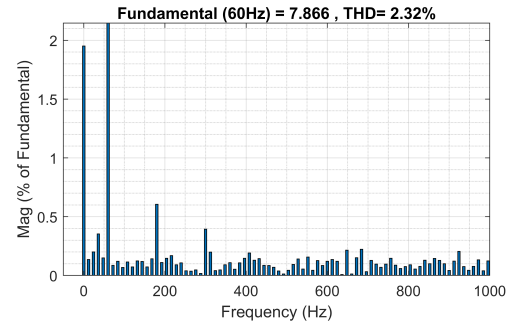

Fig. 9. The spectrum of the generated grid current i_g at 7.7A current reference peak.

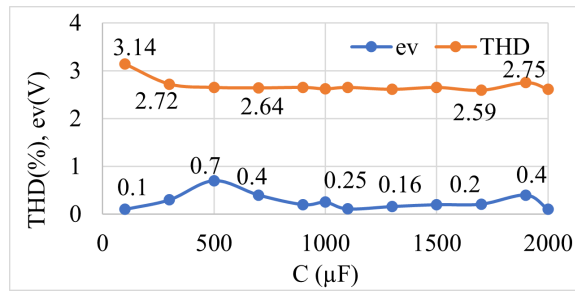

Fig. 10. The effect of varying the capacitor size on the current THD and voltage error.

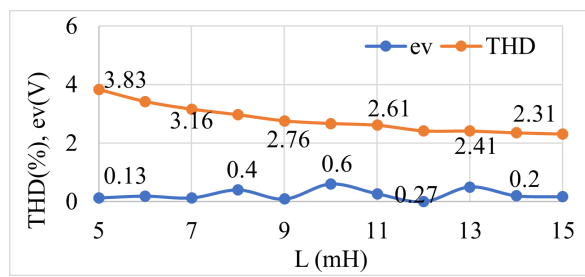

Fig. 11. The effect of varying the inductor size on the current THD and voltage error.

TABLE III
THD% OF THE GRID CURRENT AND THE MEAN SQUARE ERROR OF THE CAPACITOR VOLTAGE UNDER DIFFERENT PV CURRENT AND VOLTAGE

I_{PV} (A)	I_p^* (A)	V_1 (V)	THD% of i_g	e_V (V)
3	4.2	200	4.47	1.2
3.7	5.2	200	3.56	1.4
4.5	6.3	200	2.96	0.5
4.9	7	200	2.53	0.8
5.4	7.7	199.7	2.32	1.15
5.8	8.2	195.85	2.19	1.73
6.2	8.8	189.5	1.98	1.76

TABLE IV
PERFORMANCE COMPARISON BETWEEN THE PROPOSED RLC AND OTHER COMMON CONTROLLERS.

	PI	FCS-MPC	SMC1	SMC2	RLC
THD%	1.9	1.46	2.54	2.94	2.53
e_v (V)	7.5	0.04	0.43	0.96	1.4

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V. Conclusion

In this paper, a reinforcement learning-based controller (RLC) was proposed for a grid-connected PV-based 5-level packed-U-cell (PUC) inverter, where the control objectives are: feeding the grid with a sine-wave current with low total harmonic distortion (THD), operating at unity power factor, and maintaining the auxiliary capacitor voltage around its reference signal. The current and voltage reference values vary with the maximum power point in the PV side. The actor-critic RL architecture was employed with the proximal policy optimization (PPO) learning algorithm. The results showed that the proposed RLC coped with the dynamic nature of the problem at which the maximum absolute voltage error is less than 2V throughout the studied scenario. Moreover, the generated current signal synchronized with the grid voltage, and the THD value was 4.5% and 2% for a reference current peak that equals 4.2A and 8.8A, respectively. Furthermore, the proposed RLC showed high robustness in performance against the change in the capacitor and inductor size.

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